



A PRACTICAL POCKET GUIDE

1ST
EDITION
2026

CLINICAL CORRELATION OF LABORATORY FINDINGS

A Practical Pocket Guide for
Medical Laboratory Professionals
and Clinicians



PATTERN-BASED
INTERPRETATION



THINK OF



PATTERN
HELPS



CHECK
NEXT



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Quality Control Expert



Accurate Interpretation • Clinical Insight • Better Outcomes

Clinical Correlation of Laboratory Findings

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Professionals and Clinicians**

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Reference-Range Note

Reference intervals and decision limits vary according to laboratory, analyzer, method, age, sex, pregnancy status, altitude, and population. Always use the validated reference interval, critical-value policy, and interpretive guidance of the reporting laboratory.

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About the Author



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Mr Shankar Shahi is a Senior Medical Laboratory Technologist and Quality Control Expert with more than 20 years of experience in the medical laboratory field and more than 5 years of focused experience in Quality Control and Quality Management Systems. His professional interests include clinical laboratory interpretation, internal quality control, external quality assessment, method verification, laboratory standardization, SOP development, staff training, and practical implementation of quality systems in diagnostic laboratories. This pocket guide reflects a practical approach to translating laboratory findings into clinically meaningful patterns.

Dedicated to quality, committed to care.

Preface

Laboratory results become most valuable when they are interpreted in context rather than read as isolated numbers. This book is designed as a practical bridge between analytical laboratory work and clinical decision-making. It emphasizes specimen quality, analytical validity, reference intervals, decision limits, pattern recognition, supportive testing, critical-result awareness, and correlation with patient history.

The central approach of this pocket guide is simple: begin with the laboratory finding, **think of** the important possibilities, use the surrounding results to see how the **pattern helps**, and decide what to **check next**. The purpose is not to replace clinical diagnosis, but to help laboratory professionals and clinicians recognize meaningful laboratory patterns quickly and logically.

The intended readers include medical laboratory technologists, technicians, students, clinicians, nurses, and laboratory managers. I hope this guide serves as a practical companion at the laboratory bench, in training, and during day-to-day clinical correlation.

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How to Use This Pocket Guide

This book starts from the laboratory finding rather than from a disease chapter. Each entry gives a concise explanation of the test, a typical adult reference interval where appropriate, terminology for increased or decreased results, the main conditions to think of, useful multi-test patterns, the next tests to check, and a practical caution.

“Think of” and “Pattern Helps” indicate differential-diagnostic clues. They do not mean that a laboratory pattern alone proves a diagnosis. Clinical diagnosis requires symptoms, examination, medication and exposure history, imaging and other investigations.

Reference intervals vary by laboratory, analyzer, method, age, sex, pregnancy, altitude and population. Always use the validated reference interval, decision limit and critical-value policy of the reporting laboratory. For qualitative tests, a numeric reference range may not apply.

Critical or rapidly changing results require prompt verification and communication according to local laboratory policy and clinical urgency.

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1. Hematology

Quick-reference interpretation. Use each result as a starting point for correlation, not as a stand-alone diagnosis.

Hemoglobin (Hb)

About the Test: Hemoglobin is the oxygen-carrying protein inside red blood cells and is the principal laboratory measurement used to detect anemia or erythrocytosis. It should be interpreted with hematocrit, RBC count, red-cell indices, reticulocyte response and the clinical setting.

Reference Range: Adult men: about 13.5-17.5 g/dL; adult women: about 12.0-15.5 g/dL.

↑ Increased	↓ Decreased
Term: Erythrocytosis / polycythemia Dehydration or hemoconcentration; chronic hypoxia; smoking/high altitude; polycythemia vera; excess erythropoietin.	Term: Anemia Iron deficiency; blood loss; hemolysis; B12/folate deficiency; chronic inflammation; renal disease; marrow disorders.

Pattern Helps

- Hb low + MCV low -> microcytic anemia pattern.
- Hb low + MCV high -> macrocytic anemia pattern.
- Hb low + reticulocytes high -> blood loss or hemolysis more likely.
- Hb low + reticulocytes low/inappropriately normal -> underproduction more likely.

Clinical Correlation

- Hb low + ferritin low -> iron deficiency strongly supported.
- Hb low + indirect bilirubin/LDH high + haptoglobin low -> hemolysis strongly suspected.

Check Next: CBC indices -> reticulocyte count -> peripheral smear -> ferritin/iron studies -> B12/folate -> creatinine/eGFR; hemolysis tests when indicated.

Important Note: Hemoglobin varies with age, sex, pregnancy, altitude and hydration. A single value should not define the cause of anemia.

Hematocrit (Hct/PCV)

About the Test: Hematocrit is the proportion of blood volume occupied by red cells. It usually parallels hemoglobin and helps assess anemia, erythrocytosis and hydration status.

Reference Range: Adult men: about 41-53%; adult women: about 36-46%.

↑ Increased	↓ Decreased
Term: Elevated hematocrit Hemoconcentration; chronic hypoxia; polycythemia vera; high altitude.	Term: Low hematocrit Anemia; hemodilution; blood loss.

Pattern Helps

- Hb and Hct should broadly change together.
- Hct disproportionately high with clinical dehydration -> hemoconcentration likely.

Clinical Correlation

Hct low + Hb low + RBC count low -> anemia is supported; use MCV/RDW to classify the pattern.

Hct high + Hb high + clinical dehydration -> relative erythrocytosis from hemoconcentration is likely.

Persistent Hct/Hb elevation without dehydration -> consider chronic hypoxia or a myeloproliferative process such as polycythemia vera.

Check Next: Hb -> RBC count -> MCV/MCHC -> hydration review; repeat CBC if discordant.

Important Note: Marked Hb-Hct discordance can suggest analytical or pre-analytical problems and should trigger sample/result review.

RBC Count

About the Test: RBC count measures the number of circulating erythrocytes. It is most useful when interpreted with Hb, Hct, MCV and RDW, especially in microcytic anemia.

Reference Range: Adult men: about $4.5\text{--}5.9 \times 10^{12}/\text{L}$; adult women: about $4.1\text{--}5.1 \times 10^{12}/\text{L}$.

↑ Increased	↓ Decreased
Term: Erythrocytosis Relative hemoconcentration; chronic hypoxia; polycythemia; thalassemia trait may show a relatively high RBC count despite microcytosis.	Term: Low RBC count Most forms of anemia; blood loss; marrow suppression.

Pattern Helps

- MCV low + RBC count relatively high + RDW normal/slightly high -> thalassemia trait more likely.
- MCV low + RBC count low/normal + RDW high -> iron deficiency more likely.

Clinical Correlation

RBC count low + Hb low -> anemia is supported; classify with MCV, RDW and reticulocyte response.

MCV low + RBC count relatively preserved/high + RDW normal/slightly high -> thalassemia trait becomes more likely than iron deficiency.

RBC/Hb/Hct all high -> true or relative erythrocytosis; correlate with oxygenation, smoking/high altitude and EPO/JAK2 work-up when indicated.

Check Next: Ferritin/iron studies -> smear -> Hb analysis/HPLC when appropriate.

Important Note: RBC count alone does not define anemia; hemoglobin remains the key concentration measure.

MCV

About the Test: Mean corpuscular volume represents average red-cell size and is central to classifying anemia as microcytic, normocytic or macrocytic.

Reference Range: About 80-100 fL in adults.

↑ Increased	↓ Decreased
Term: Macrocytosis B12 or folate deficiency; alcohol use; liver disease; hypothyroidism; reticulocytosis; marrow disorders;	Term: Microcytosis Iron deficiency; thalassemia; chronic inflammation; sideroblastic anemia; lead-related disorders.

selected drugs.

Pattern Helps

- MCV low + ferritin low -> iron deficiency likely.
- MCV low + ferritin normal/high + RBC relatively high -> consider thalassemia trait.
- MCV high + B12 low -> B12 deficiency likely.

Clinical Correlation

MCV low + ferritin low -> iron deficiency is strongly supported.

MCV low + ferritin normal/high + RBC count relatively high -> consider thalassemia trait.

MCV high + B12/folate low -> megaloblastic anemia is likely; macrocytosis with normal vitamins should prompt liver, thyroid, alcohol/drug and marrow review.

Check Next: RDW -> ferritin/iron profile -> smear -> reticulocytes -> B12/folate -> TSH/liver tests as indicated.

Important Note: Mixed deficiencies can produce a near-normal MCV; RDW and smear can reveal mixed cell populations.

MCH / MCHC

About the Test: MCH reflects average hemoglobin content per red cell; MCHC reflects hemoglobin concentration within red cells. They support characterization of hypochromia and identify unusual patterns such as unexpectedly high MCHC.

Reference Range: MCH: about 27-33 pg; MCHC: about 32-36 g/dL.

↑ Increased

Term: High MCHC

Uncommon; hereditary spherocytosis or analytical interference/artifact should be considered.

↓ Decreased

Term: Low MCH/MCHC

Hypochromia, most often with iron deficiency or thalassemia.

Pattern Helps

- Low MCV + low MCH/MCHC -> hypochromic microcytic pattern.
- Unexpectedly high MCHC -> review smear and specimen for hemolysis, lipemia, agglutination or analyzer flags.

Clinical Correlation

Low MCH/MCHC + low MCV + low ferritin -> hypochromic microcytic iron-deficiency pattern.

Low MCH/MCHC + low MCV + relatively high RBC count -> thalassemia trait should be considered.

Unexpectedly high MCHC + spherocytes -> hereditary spherocytosis is possible; marked isolated elevation also warrants artifact/sample review.

Check Next: Smear -> ferritin/iron studies -> repeat/sample review if implausible.

Important Note: MCH and MCHC are supportive indices and should not be interpreted in isolation.

RDW

About the Test: Red cell distribution width measures variation in red-cell size (anisocytosis). It is useful in differentiating causes of microcytosis and recognizing mixed red-cell populations.

Reference Range: Commonly about 11.5-14.5% (method dependent).

↑ Increased	↓ Decreased
Term: Anisocytosis / elevated RDW Iron deficiency; mixed nutritional deficiency; recent transfusion; reticulocytosis/recovery; mixed anemia.	Low RDW is usually not clinically important.

Pattern Helps

- Low MCV + high RDW -> iron deficiency more likely.
- Low MCV + normal RDW + relatively high RBC count -> thalassemia trait more likely.
- High MCV + high RDW -> B12/folate deficiency or mixed process.

Clinical Correlation

High RDW + low MCV + low ferritin -> evolving or established iron deficiency is likely.

Normal RDW + marked microcytosis + relatively high RBC count -> thalassemia trait is favored.

High RDW + macrocytosis -> mixed nutritional deficiency, reticulocytosis or marrow stress should be considered.

Check Next: MCV -> ferritin/iron studies -> smear -> B12/folate when indicated.

Important Note: RDW supports pattern recognition but is not diagnostic by itself.

WBC Count

About the Test: Total white blood cell count measures circulating leukocytes. The differential count and absolute cell counts are essential for meaningful interpretation.

Reference Range: About $4.0\text{--}11.0 \times 10^9/\text{L}$ in adults.

↑ Increased	↓ Decreased
Term: Leukocytosis Infection; inflammation; physiologic stress; corticosteroids; tissue injury; smoking; myeloproliferative or leukemic disorders.	Term: Leukopenia Viral illness; severe sepsis; marrow suppression/failure; drugs/chemotherapy; autoimmune disease; hypersplenism.

Pattern Helps

- WBC high + neutrophilia -> bacterial/inflammatory/stress pattern.
- WBC high + lymphocytosis -> viral or lymphoproliferative pattern.
- WBC low + neutropenia -> increased infection risk depending on ANC.

Clinical Correlation

WBC high + neutrophilia + CRP/PCT high -> bacterial infection or acute inflammation becomes more likely.

WBC high + lymphocytosis -> viral infection is common; persistent marked lymphocytosis may require evaluation for a lymphoproliferative disorder.

WBC low + neutropenia -> consider viral illness, drug effect, marrow suppression or severe systemic illness.

Check Next: Differential count -> absolute counts -> smear -> CRP/procalcitonin/cultures as indicated; hematology evaluation if persistent/unexplained.

Important Note: Use age-appropriate ranges. Automated flags or unexplained abnormalities should prompt peripheral smear review.

Neutrophils / ANC

About the Test: Neutrophils are major innate immune cells. Absolute neutrophil count (ANC) is more clinically useful than percentage alone, especially for assessing neutropenia.

Reference Range: ANC about $1.5-7.5 \times 10^9/L$ in adults.

↑ Increased	↓ Decreased
Term: Neutrophilia Bacterial infection; inflammation; tissue necrosis; corticosteroids/stress; smoking; myeloproliferative disease.	Term: Neutropenia Viral infection; drugs/chemotherapy; marrow failure; severe sepsis; autoimmune neutropenia; nutritional deficiency.

Pattern Helps

- Neutrophilia + left shift/toxic change -> supports acute bacterial/inflammatory response.
- ANC $<0.5 \times 10^9/L$ -> severe neutropenia with high infection risk.

Clinical Correlation

Neutrophilia + left shift/toxic change + raised inflammatory markers -> acute bacterial infection is strongly supported.

Neutropenia + fever -> febrile neutropenia is a medical emergency and requires urgent clinical assessment.

Neutropenia with pancytopenia -> marrow suppression/failure, infiltrative disease, drugs or severe infection should be considered.

Check Next: Peripheral smear -> infection work-up/cultures as indicated -> medication review -> B12/folate; marrow assessment if persistent.

Important Note: Percentages can mislead when total WBC is abnormal; use absolute neutrophil count.

Lymphocytes

About the Test: Lymphocytes include T, B and NK cells. Absolute lymphocyte count is preferred to percentage alone when assessing lymphocytosis or lymphopenia.

Reference Range: About $1.0\text{--}4.0 \times 10^9/\text{L}$ in adults.

↑ Increased	↓ Decreased
Term: Lymphocytosis Many viral infections; pertussis; some chronic infections; smoking; lymphoproliferative disorders.	Term: Lymphopenia Corticosteroids; severe illness; immunodeficiency; chemotherapy; some viral infections.

Pattern Helps

- Reactive/atypical lymphocytes + compatible illness -> viral/reactive process.
- Persistent marked monomorphic lymphocytosis -> consider lymphoproliferative disease.

Clinical Correlation

Absolute lymphocytosis + atypical/reactive lymphocytes -> viral infection such as EBV/CMV is likely in the appropriate context.

Persistent absolute lymphocytosis with monomorphic cells -> consider a lymphoproliferative disorder and flow cytometry.

Lymphopenia + immunosuppression/systemic illness -> consider viral infection, steroid effect or immune deficiency depending on context.

Check Next: Peripheral smear -> targeted viral testing -> flow cytometry if persistent/unexplained.

Important Note: Age strongly affects normal lymphocyte counts, especially in children.

Monocytes

About the Test: Monocytes participate in phagocytosis and chronic inflammatory responses. Mild monocytosis is common and non-specific; persistent monocytosis may require further evaluation.

Reference Range: About 0.2-0.8 x10⁹/L.

↑ Increased	↓ Decreased
Term: Monocytosis Recovery from acute infection; chronic infections; inflammatory/autoimmune disorders; selected myeloid neoplasms.	Term: Monocytopenia Usually non-specific; may occur with marrow suppression or corticosteroid exposure.

Pattern Helps

- Persistent monocytosis + cytopenias/dysplasia -> consider myeloid disorder.

Clinical Correlation

Persistent monocytosis + inflammatory symptoms -> chronic infection/inflammation should be considered.

Monocytosis with cytopenias or dysplastic cells -> consider a myeloid neoplasm such as CMML and obtain hematology review.

Transient monocytosis during recovery from neutropenia/infection may represent marrow recovery.

Check Next: Repeat CBC -> smear -> inflammatory/infectious evaluation -> hematology work-up if persistent.

Important Note: Interpret absolute monocyte count rather than percentage alone.

Eosinophils

About the Test: Eosinophils are involved in allergic inflammation, parasite responses and selected drug, autoimmune and hematologic conditions.

Reference Range: About 0.0-0.5 x10⁹/L.

↑ Increased	↓ Decreased
Term: Eosinophilia Allergic disease; parasitic infection;	Term: Eosinopenia Usually not clinically significant; can

drug reaction; eosinophilic disorders; occur with stress or corticosteroids.
some autoimmune and hematologic diseases.

Pattern Helps

- Eosinophilia + travel/exposure/GI symptoms -> consider parasitic infection.
- Eosinophilia + rash/new drug -> consider drug hypersensitivity.

Clinical Correlation

Eosinophilia + asthma/allergic symptoms -> allergic disease is likely.

Eosinophilia + travel/exposure/GI symptoms -> parasitic infection should be considered.

Marked persistent eosinophilia with organ involvement -> consider drug reaction, hypereosinophilic syndrome or hematologic disease.

Check Next: Absolute eosinophil count -> exposure/travel/drug review -> parasite testing or allergy evaluation as indicated.

Important Note: Persistent marked eosinophilia deserves systematic evaluation for organ involvement and clonal causes.

Basophils

About the Test: Basophils are a small granulocyte population involved in hypersensitivity responses. Persistent basophilia is uncommon and can be a clue to myeloproliferative disease.

Reference Range: About $0.0-0.1 \times 10^9/L$.

↑ Increased	↓ Decreased
Term: Basophilia Myeloproliferative neoplasms, especially CML; allergy; chronic inflammation; hypothyroidism in some cases.	Term: Basopenia Usually not clinically significant.

Pattern Helps

- Basophilia + marked leukocytosis + left shift -> consider CML/myeloproliferative process.

Clinical Correlation

Persistent basophilia + marked leukocytosis + left shift -> chronic myeloid leukemia should be considered.

Basophilia with other myeloproliferative features -> evaluate for a myeloproliferative neoplasm.

Mild isolated basophilia can occur with allergy/inflammation and should be interpreted with the full CBC and smear.

Check Next: Repeat CBC -> smear -> hematology evaluation; BCR-ABL1 testing when clinically appropriate.

Important Note: Small absolute numbers make percentage changes easy to overinterpret.

Platelet Count

About the Test: Platelets are central to primary hemostasis. The count helps identify thrombocytopenia or thrombocytosis, but function may be abnormal despite a normal count.

Reference Range: About $150-450 \times 10^9/L$.

↑ Increased	↓ Decreased
Term: Thrombocytosis Reactive inflammation/infection; iron deficiency; tissue injury; postsplenectomy; malignancy; myeloproliferative neoplasm.	Term: Thrombocytopenia Immune destruction; drugs; infection/sepsis; DIC; liver disease/hypersplenism; marrow failure; dilution; pseudothrombocytopenia.

Pattern Helps

- Platelets low + PT/aPTT prolonged + D-dimer high + fibrinogen low -> DIC/consumption pattern.
- Platelets high + ferritin low -> reactive thrombocytosis from iron deficiency possible.
- Isolated low platelets with clumping on smear -> pseudothrombocytopenia.

Clinical Correlation

Platelets low + prolonged PT/aPTT + high D-dimer + low fibrinogen -> DIC is strongly suspected.

Platelets low + normal PT/aPTT + mucocutaneous bleeding -> an isolated platelet disorder such as ITP becomes more likely.

Platelets high + low ferritin -> reactive thrombocytosis due to iron deficiency is common; persistent unexplained thrombocytosis warrants MPN evaluation.

Check Next: Repeat count/smear -> citrate sample if clumping -> PT/aPTT/fibrinogen/D-dimer -> liver tests -> ferritin/CRP; hematology review if persistent.

Important Note: Always exclude platelet clumping before labeling unexpected thrombocytopenia.

Reticulocyte Count

About the Test: Reticulocytes are young red cells released from marrow. They indicate whether marrow is appropriately responding to anemia. A corrected reticulocyte count or reticulocyte production index is more useful in significant anemia.

Reference Range: About 0.5-2.5% in adults; absolute count is preferred when available.

↑ Increased	↓ Decreased
Term: Reticulocytosis Hemolysis; recent blood loss; recovery after treatment of deficiency.	Term: Reticulocytopenia / inadequate response Marrow underproduction; untreated iron/B12/folate deficiency; renal disease; marrow suppression/failure.

Pattern Helps

- Anemia + reticulocytes high -> blood loss or hemolysis.
- Anemia + reticulocytes low/inappropriately normal -> production problem.

Clinical Correlation

Anemia + reticulocytes high -> marrow response is present; think blood loss or hemolysis.

Anemia + reticulocytes low/inappropriately normal -> underproduction from iron/B12/folate deficiency, renal disease or marrow failure is more likely.

Reticulocytosis + indirect bilirubin/LDH high + haptoglobin low -> hemolysis is strongly supported.

Check Next: LDH -> bilirubin -> haptoglobin -> smear -> iron/B12/folate -> creatinine/eGFR as indicated.

Important Note: Interpret relative to the severity of anemia; a “normal” percentage may be inadequate when Hb is low.

Peripheral Blood Smear

About the Test: Peripheral smear provides direct morphological assessment of red cells, white cells and platelets. It complements automated CBC results and is essential when analyzer flags or unusual patterns are present.

Reference Range: No single numeric reference range; morphology should be predominantly normal for age and clinical context.

↑ Increased	↓ Decreased
Term: Abnormal morphology No specific clinically useful high-state terminology.	No specific clinically useful low-state interpretation.

Pattern Helps

- Schistocytes -> microangiopathic hemolysis/DIC/TMA should be considered.
- Spherocytes -> hereditary spherocytosis or immune hemolysis.
- Target cells -> thalassemia, liver disease, hemoglobinopathy or postsplenectomy.
- Blasts -> acute leukemia or severe marrow disorder until evaluated.

Clinical Correlation

Schistocytes + thrombocytopenia + hemolysis markers -> microangiopathic hemolysis/DIC/TMA should be urgently considered.

Spherocytes + hemolysis -> hereditary spherocytosis or immune hemolysis; correlate with DAT where appropriate.

Blasts or highly abnormal immature cells -> acute leukemia or marrow disease must be considered and requires urgent hematology evaluation.

Microcytosis/hypochromia + pencil cells -> iron deficiency; target cells with microcytosis may suggest thalassemia/hemoglobinopathy.

Check Next: Correlate with CBC/reticulocytes; order hemolysis tests, Hb analysis, flow cytometry or marrow studies according to morphology.

Important Note: Smear interpretation requires trained review and should be integrated with automated flags and clinical information.

ESR

About the Test: Erythrocyte sedimentation rate is a non-specific marker influenced by inflammation, immunoglobulins, anemia, age and red-cell morphology. It changes more slowly than CRP.

Reference Range: Commonly <15 mm/h in younger men and <20 mm/h in younger women; ranges rise with age and method.

↑ Increased	↓ Decreased
Term: Elevated ESR Inflammation; autoimmune disease; infection; anemia; plasma-cell disorders; pregnancy; malignancy.	Term: Low ESR Polycythemia; marked leukocytosis; sickle/abnormal RBC shapes; often not clinically important.

Pattern Helps

- ESR high + CRP high -> active inflammatory process more likely.
- Very high ESR + high total protein/globulin -> consider monoclonal/inflammatory protein disorders.

Clinical Correlation

ESR high + CRP high -> active inflammation/infection is supported.

ESR very high + anemia/renal dysfunction/high globulin -> consider plasma-cell dyscrasia or chronic inflammatory disease.

ESR high with relatively normal CRP can occur with anemia, pregnancy, autoimmune disease or paraproteinemia; interpret cautiously.

Check Next: CRP -> CBC -> targeted autoimmune/infectious testing -> serum protein electrophoresis if clinically indicated.

Important Note: ESR is non-specific and should not be used alone to diagnose or exclude disease.

Hemoglobin Analysis / HPLC

About the Test: Hemoglobin electrophoresis or HPLC separates hemoglobin fractions and supports diagnosis of hemoglobin variants and thalassemias. Interpretation depends on age, transfusion history and the analytic platform.

Reference Range: Adult HbA usually ~95-98%, HbA2 ~2-3.5%, HbF usually <1-2% (method dependent).

↑ Increased	↓ Decreased
Term: Abnormal hemoglobin fraction No specific clinically useful high-state terminology.	No specific clinically useful low-state interpretation.

Pattern Helps

- Microcytosis + normal ferritin + HbA2 elevated -> beta-thalassemia trait likely.
- HbS detected -> sickle hemoglobin present; quantify fractions and correlate with genotype/transfusion status.

Clinical Correlation

Microcytosis + normal ferritin + elevated HbA2 -> beta-thalassemia trait is strongly suggested.

Abnormal Hb variant peak + compatible CBC/smear -> hemoglobinopathy should be considered and confirmed according to method.

Low MCV with normal HbA2 does not exclude alpha-thalassemia; genetic testing may be needed when clinically important.

Check Next: CBC/MCV -> ferritin -> family history -> molecular testing when needed.

Important Note: Recent transfusion can significantly alter hemoglobin fraction patterns.

G6PD Activity

About the Test: G6PD protects red cells from oxidative injury. Deficiency predisposes to episodic hemolysis after oxidant drugs, fava beans or infection.

Reference Range: Assay-specific; commonly reported as activity per gram Hb with a laboratory-defined lower limit.

↑ Increased	↓ Decreased
High activity is generally not clinically important.	Term: G6PD deficiency Inherited enzyme deficiency; may be missed during acute hemolysis because older deficient cells have already been destroyed.

Pattern Helps

- Hemolysis after oxidant exposure + low G6PD activity -> G6PD deficiency strongly supported.

Clinical Correlation

Hemolysis after oxidant drug/infection + low G6PD activity -> G6PD deficiency is strongly supported.

Heinz body/bite-cell pattern with hemolysis -> oxidant injury including G6PD deficiency should be considered.

Normal G6PD during an acute hemolytic episode does not fully exclude deficiency because young reticulocytes may raise measured activity.

Check Next: CBC -> reticulocytes -> bilirubin/LDH/haptoglobin -> repeat G6PD after recovery if initial test is normal but suspicion remains high.

Important Note: Testing during or soon after acute hemolysis can give a false-normal result due to reticulocytosis.

2. Coagulation

Quick-reference interpretation. Use each result as a starting point for correlation, not as a stand-alone diagnosis.

PT / INR

About the Test: Prothrombin time assesses the extrinsic and common coagulation pathways. INR standardizes PT primarily for vitamin K antagonist monitoring.

Reference Range: PT often ~11-14 s; INR ~0.8-1.2 in people not receiving warfarin (reagent dependent).

↑ Increased	↓ Decreased
Term: Prolonged PT/INR Warfarin effect; vitamin K deficiency; liver synthetic dysfunction; DIC; factor VII/common-pathway factor deficiency; massive transfusion.	Short PT is usually not clinically meaningful.

Pattern Helps

- PT prolonged + aPTT normal -> factor VII deficiency/inhibitor, early vitamin K deficiency or warfarin effect.
- PT + aPTT prolonged -> common pathway defect, severe liver disease, DIC or anticoagulant effect.

Clinical Correlation

PT/INR prolonged + normal aPTT -> factor VII deficiency, early vitamin K deficiency/warfarin effect or early liver dysfunction should be considered.

PT and aPTT both prolonged + low fibrinogen/high D-dimer/low platelets -> DIC is strongly suspected.

Prolonged INR + low albumin or other liver dysfunction markers -> impaired hepatic synthetic function becomes more likely.

Check Next: aPTT -> fibrinogen -> D-dimer -> liver tests -> mixing study/factor assays if unexplained.

Important Note: INR should not be used to standardize PT for every indication; it is designed for vitamin K antagonist monitoring.

aPTT

About the Test: Activated partial thromboplastin time assesses intrinsic and common pathways and is used in selected settings for unfractionated heparin monitoring.

Reference Range: Often ~25-35 s, but strongly reagent dependent.

↑ Increased	↓ Decreased
Term: Prolonged aPTT Unfractionated heparin; lupus anticoagulant; factor VIII/IX/XI/XII deficiency; acquired inhibitors; DIC; severe liver disease.	Short aPTT is usually non-specific; may occur with high factor VIII/inflammation.

Pattern Helps

- aPTT prolonged + PT normal -> intrinsic factor deficiency, inhibitor or heparin effect.
- Correction on mixing study -> factor deficiency more likely; no correction -> inhibitor/anticoagulant more likely.

Clinical Correlation

Isolated prolonged aPTT + bleeding -> intrinsic factor deficiency or acquired inhibitor should be considered.

Isolated prolonged aPTT without bleeding/thrombosis history -> lupus anticoagulant or heparin contamination/effect may be possible.

PT and aPTT both prolonged -> consider DIC, severe liver disease, massive transfusion or common-pathway factor deficiency.

Check Next: PT -> thrombin time/anti-Xa if anticoagulant suspected -> mixing study -> factor assays/lupus anticoagulant evaluation.

Important Note: Pre-analytical errors such as underfilled citrate tubes can falsely prolong clotting times.

Fibrinogen

About the Test: Fibrinogen is a liver-synthesized clotting protein and an acute-phase reactant. It is consumed in severe disseminated coagulation and may rise with inflammation.

Reference Range: About 200-400 mg/dL (2.0-4.0 g/L).

↑ Increased	↓ Decreased
Term: Hyperfibrinogenemia Inflammation; infection; pregnancy; tissue injury; cardiovascular risk states.	Term: Hypofibrinogenemia DIC/consumption; severe liver disease; massive hemorrhage/transfusion; congenital deficiency.

Pattern Helps

- Platelets low + D-dimer high + fibrinogen low/decreasing -> DIC pattern.

Clinical Correlation

Low fibrinogen + thrombocytopenia + prolonged PT/aPTT + high D-dimer -> consumptive coagulopathy/DIC is strongly supported.

High fibrinogen + high CRP/ESR -> acute-phase inflammatory response is likely.

Low fibrinogen in severe liver disease may reflect impaired synthesis or consumption.

Check Next: PT/aPTT -> D-dimer -> platelet count -> liver tests; repeat serially if DIC suspected.

Important Note: Fibrinogen can remain normal early in DIC because it is also an acute-phase reactant.

D-dimer

About the Test: D-dimer is produced when cross-linked fibrin is degraded. A negative high-sensitivity test can help exclude venous thromboembolism in appropriately selected low/intermediate-risk patients, but a positive result is non-specific.

Reference Range: Assay- and unit-specific; many FEU assays use <0.50 mg/L FEU (500 ng/mL FEU) as a conventional cutoff.

↑ Increased	↓ Decreased
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Term: Elevated D-dimer
VTE; DIC; infection/inflammation;
cancer; pregnancy; recent
surgery/trauma; older age.

A low/negative result is mainly useful
for exclusion in appropriate clinical
probability settings.

Pattern Helps

- D-dimer high + platelets low + PT prolonged + fibrinogen low -> DIC strongly considered.

Clinical Correlation

High D-dimer + high clinical probability of VTE -> venous thromboembolism requires definitive imaging; D-dimer alone does not diagnose VTE.

High D-dimer + low platelets + prolonged PT/aPTT + low fibrinogen -> DIC should be considered.

Normal age/assay-appropriate D-dimer in a validated low-risk pathway can help exclude VTE.

Check Next: Apply clinical probability -> imaging for VTE when indicated; PT/aPTT/fibrinogen/platelets if DIC suspected.

Important Note: Do not diagnose thrombosis from D-dimer alone. Age-adjusted thresholds may be used in some validated pathways.

Thrombin Time (TT)

About the Test: Thrombin time measures conversion of fibrinogen to fibrin after addition of thrombin. It is sensitive to heparin, direct thrombin inhibitors and major fibrinogen abnormalities.

Reference Range: Often ~14-21 s, method dependent.

↑ Increased	↓ Decreased
Term: Prolonged TT Heparin; direct thrombin inhibitors; hypofibrinogenemia; dysfibrinogenemia; high fibrin degradation products.	Short TT is generally not clinically significant.

Pattern Helps

- TT prolonged with normal reptilase time -> heparin or direct thrombin inhibitor more likely.

Clinical Correlation

Prolonged TT + normal reptilase time -> heparin/direct thrombin inhibitor effect is likely.

Prolonged TT + prolonged reptilase time -> low/abnormal fibrinogen or fibrin degradation products should be considered.

Unexpected TT prolongation from a line-drawn sample -> heparin contamination is a common pre-analytical explanation.

Check Next: Fibrinogen -> medication/anticoagulant review -> reptilase time or specific drug testing if available.

Important Note: TT is very sensitive to heparin contamination, including samples drawn from heparinized lines.

3. Liver & Hepatobiliary

Quick-reference interpretation. Use each result as a starting point for correlation, not as a stand-alone diagnosis.

Total, Direct & Indirect Bilirubin

About the Test: Bilirubin is a yellow pigment formed mainly from heme breakdown. Unconjugated bilirubin is transported to the liver, conjugated by hepatocytes and excreted into bile. Fractionation into total and direct bilirubin helps distinguish hemolytic/conjugation disorders from hepatocellular or cholestatic disease.

Reference Range: Total bilirubin: about 0.3-1.2 mg/dL; direct bilirubin: about 0.0-0.3 mg/dL. Indirect bilirubin is commonly estimated as total minus direct.

↑ Increased	↓ Decreased
Term: Hyperbilirubinemia Hemolysis; Gilbert syndrome; ineffective erythropoiesis; hepatitis; drug-induced liver injury; cholestasis; biliary obstruction; advanced liver dysfunction.	Low bilirubin is usually not clinically significant.

Pattern Helps

- Total bilirubin high + direct normal/low -> predominantly unconjugated hyperbilirubinemia.
- Direct bilirubin high + ALP/GGT high -> cholestatic/obstructive pattern.
- AST/ALT markedly high + bilirubin high -> hepatocellular injury pattern.

Clinical Correlation

- Bilirubin high + LDH high + reticulocytes high + haptoglobin low -> hemolysis strongly suspected.
- Mild isolated indirect bilirubin high + AST/ALT normal + hemolysis markers normal -> Gilbert syndrome should be considered.
- Direct bilirubin high + ALP/GGT disproportionately high -> cholestasis/biliary obstruction more likely.

Check Next: CBC -> reticulocytes -> LDH -> haptoglobin -> smear when unconjugated; AST/ALT -> ALP -> GGT -> PT/INR -> albumin -> imaging when conjugated/cholestatic.

Important Note: Interpret bilirubin with fractionation, liver enzymes and hemolysis markers. A single elevated value does not establish a diagnosis.

AST (SGOT)

About the Test: AST is an aminotransferase found in liver, skeletal muscle, heart and other tissues. It is less liver-specific than ALT, so an isolated AST rise should prompt consideration of muscle and other sources.

Reference Range: Common adult range about 10-40 U/L, method dependent.

↑ Increased	↓ Decreased
Term: Elevated AST Hepatocellular injury; alcohol-related liver injury; muscle injury/rhabdomyolysis; hemolysis; strenuous exercise; myocardial injury less commonly used diagnostically now.	Low AST is usually not clinically important.

Pattern Helps

- AST high + ALT high -> hepatocellular injury pattern.
- AST disproportionately higher than ALT -> consider alcohol-related liver disease, advanced fibrosis/cirrhosis or muscle source depending on context.
- AST high + CK markedly high -> skeletal muscle injury more likely.

Clinical Correlation

AST high + ALT high with ALT predominance -> hepatocellular injury is likely.

AST disproportionately high + history of alcohol use + elevated GGT -> alcohol-associated liver injury may be considered.

AST high + CK markedly high with relatively lesser ALT rise -> skeletal muscle injury is more likely than primary liver disease.

Check Next: ALT -> ALP/GGT -> bilirubin -> CK if muscle source possible -> PT/INR/albumin if significant liver disease.

Important Note: Hemolyzed samples can increase AST. AST/ALT ratio is supportive, not diagnostic.

ALT (SGPT)

About the Test: ALT is an aminotransferase concentrated mainly in hepatocytes and is generally more liver-specific than AST. Elevation indicates hepatocellular injury rather than hepatic function.

Reference Range: Common adult range about 7-45 U/L, method/sex dependent.

↑ Increased	↓ Decreased
Term: Elevated ALT Viral hepatitis; metabolic dysfunction-associated steatotic liver disease; drug/toxin injury; ischemic injury; autoimmune hepatitis; other hepatocellular disorders.	Low ALT is generally not clinically significant.

Pattern Helps

- ALT predominant with AST elevation -> common hepatocellular pattern.
- ALT/AST high with relatively normal ALP -> hepatocellular rather than cholestatic pattern.

Clinical Correlation

ALT high + AST high with normal/mild ALP -> hepatocellular injury pattern; consider viral, metabolic, drug-related or ischemic causes according to magnitude/context.

ALT/AST very high + INR prolonged -> severe acute hepatocellular injury requires urgent assessment.

Persistent mild ALT elevation + metabolic risk factors -> metabolic dysfunction-associated steatotic liver disease should be considered.

Check Next: AST -> ALP/GGT -> bilirubin -> hepatitis testing/medication review -> imaging and further liver work-up according to context.

Important Note: ALT may be normal in some chronic liver disease. “Normal” ranges differ by laboratory and population.

Alkaline Phosphatase (ALP)

About the Test: ALP is produced mainly by biliary epithelium and bone. Elevation therefore requires determination of whether the source is hepatobiliary or non-hepatic.

Reference Range: About 40-130 U/L in adults; higher values are normal in growing children and may rise in pregnancy.

↑ Increased	↓ Decreased
Term: Elevated ALP Biliary obstruction/cholestasis; infiltrative liver disease; bone growth/turnover; fractures; Paget disease; bone metastases; pregnancy.	Term: Low ALP Malnutrition; zinc/magnesium deficiency; hypothyroidism; hypophosphatasia (rare).

Pattern Helps

- ALP high + GGT high -> hepatobiliary source favored.
- ALP high + GGT normal -> consider bone or other non-hepatic source.
- ALP high + direct bilirubin high -> obstructive/cholestatic process more likely.

Clinical Correlation

ALP high + GGT high + direct bilirubin high -> hepatobiliary cholestasis/obstruction is strongly supported.

ALP high + GGT normal + calcium/phosphate/PTH abnormalities -> a bone source becomes more likely.

ALP high in growing children/pregnancy can be physiologic; interpret using age/pregnancy-specific ranges.

Check Next: GGT -> bilirubin -> AST/ALT -> ALP isoenzyme or bone profile when needed -> imaging if hepatobiliary pattern.

Important Note: Age, pregnancy and bone growth strongly affect ALP; use age- and sex-appropriate ranges.

Gamma-Glutamyl Transferase (GGT)

About the Test: GGT is a sensitive but non-specific enzyme used mainly to support a hepatobiliary source of elevated ALP. It can also rise with alcohol exposure and enzyme-inducing drugs.

Reference Range: Commonly about 9-48 U/L; sex and method dependent.

↑ Increased	↓ Decreased
Term: Elevated GGT Cholestasis/biliary disease; alcohol exposure; fatty liver; medications/enzyme induction; other liver injury.	Low GGT is not clinically significant.

Pattern Helps

- ALP high + GGT high -> hepatobiliary source likely.
- ALP high + GGT normal -> bone source becomes more likely.

Clinical Correlation

GGT high + ALP high -> hepatic/biliary source of ALP elevation is supported.

ALP high + GGT normal -> consider bone rather than hepatobiliary disease.

Isolated GGT elevation may reflect alcohol use, enzyme-inducing drugs or metabolic liver disease and is nonspecific.

Check Next: ALP -> bilirubin -> AST/ALT -> medication/alcohol history -> imaging when cholestasis suspected.

Important Note: GGT is sensitive but poorly specific; it should not be used alone to diagnose alcohol-related liver disease.

Albumin

About the Test: Albumin is the major plasma protein synthesized by the liver. It reflects liver synthetic capacity but is also strongly affected by inflammation, nutrition, renal/GI loss and fluid status.

Reference Range: About 3.5-5.0 g/dL.

↑ Increased	↓ Decreased
Term: Hyperalbuminemia	Term: Hypoalbuminemia

Usually
dehydration/hemoconcentration.

Chronic liver disease; nephrotic protein
loss; inflammation; malnutrition;
protein-losing enteropathy; burns;
dilution.

Pattern Helps

- Albumin low + urine protein heavy -> renal protein loss/nephrotic pattern.
- Albumin low + INR prolonged + chronic liver findings -> impaired hepatic synthetic function.

Clinical Correlation

Albumin low + edema + heavy proteinuria -> nephrotic protein loss should be considered.

Albumin low + prolonged INR + chronic liver abnormalities -> impaired hepatic synthetic function is likely.

Albumin low + high CRP -> inflammation may contribute; albumin alone is not a specific nutritional marker.

Check Next: Total protein/globulin -> urine protein/ACR -> liver tests/PT-INR -> CRP -> nutritional/GI evaluation as indicated.

Important Note: Albumin is a negative acute-phase reactant; low levels do not necessarily mean poor nutrition.

Total Protein / Globulin / A:G Ratio

About the Test: Total protein is the combined concentration of albumin and globulins. Calculated globulin and the albumin-to-globulin ratio help detect inflammatory, immune or monoclonal protein patterns.

Reference Range: Total protein about 6.0-8.3 g/dL; albumin/globulin ratio often ~1.0-2.5.

↑ Increased

Term: Hyperproteinemia /
hyperglobulinemia
Dehydration; chronic
inflammation/infection; autoimmune
disease; monoclonal gammopathy.

↓ Decreased

Term: Hypoproteinemia
Protein loss; liver disease; malnutrition;
immunodeficiency; dilution.

Pattern Helps

- Total protein high + albumin normal/low + globulin high -> inflammatory or monoclonal protein process.
- Low A:G ratio + high globulin -> chronic inflammation, liver disease or gammopathy.

Clinical Correlation

Total protein/globulin high + low A:G ratio + anemia/renal dysfunction -> monoclonal gammopathy or chronic inflammation should be considered.

Total protein low + albumin low -> reduced synthesis, protein loss, malnutrition/malabsorption or dilution may be present.

High globulin + autoimmune/infectious features -> polyclonal hypergammaglobulinemia is possible; SPEP helps distinguish patterns.

Check Next: Albumin -> SPEP/immunofixation/free light chains if indicated -> CRP/ESR -> liver/renal evaluation.

Important Note: A monoclonal gammopathy cannot be diagnosed from total protein or A:G ratio alone.

4. Renal Function & Electrolytes

Quick-reference interpretation. Use each result as a starting point for correlation, not as a stand-alone diagnosis.

Urea / BUN

About the Test: Urea is a nitrogenous waste product formed from protein metabolism and excreted by the kidneys. It is influenced by renal function, hydration, protein intake, catabolism and gastrointestinal bleeding.

Reference Range: Urea: about 15-40 mg/dL (2.5-6.7 mmol/L); BUN: about 7-20 mg/dL.

↑ Increased	↓ Decreased
Term: Azotemia / elevated urea Reduced GFR; dehydration/prerenal state; GI bleeding; high protein intake; catabolic state; corticosteroids.	Term: Low urea Low protein intake; severe liver dysfunction; overhydration; pregnancy.

Pattern Helps

- Urea disproportionately high relative to creatinine -> prerenal state, GI bleed or high catabolism possible.
- Urea high + creatinine high -> renal impairment or prerenal/postrenal process depending on context.

Clinical Correlation

Urea/BUN disproportionately high relative to creatinine + dehydration signs -> prerenal azotemia is likely.

Urea/BUN high + GI bleeding features -> increased protein absorption from upper GI bleeding should be considered.

Urea/BUN low with severe liver dysfunction or low protein intake may reflect reduced urea production/intake.

Check Next: Creatinine/eGFR -> electrolytes -> urinalysis -> volume status -> renal imaging when obstruction suspected.

Important Note: Urea is less specific for GFR than creatinine/eGFR.

Creatinine

About the Test: Creatinine is produced from muscle creatine and is used to estimate GFR. Its concentration is affected by muscle mass, diet, age and selected drugs, so eGFR is usually more informative than creatinine alone.

Reference Range: About 0.6-1.3 mg/dL in adults, varying with sex, muscle mass and method.

↑ Increased	↓ Decreased
Term: Elevated creatinine / azotemia Acute kidney injury; chronic kidney disease; dehydration/prerenal state; obstruction; increased muscle breakdown; some drugs that alter tubular secretion.	Term: Low creatinine Low muscle mass; pregnancy; severe muscle wasting; overhydration.

Pattern Helps

- Creatinine rising rapidly -> acute kidney injury pattern.
- Creatinine high + eGFR low + persistent albuminuria -> chronic kidney disease more likely if sustained >=3 months.
- Creatinine high + CK high -> consider rhabdomyolysis-related kidney injury.

Clinical Correlation

Creatinine rising + eGFR falling -> declining kidney filtration is strongly supported.

Creatinine high + hyperkalemia + bicarbonate low -> clinically significant renal dysfunction/AKI pattern should be considered.

Creatinine rise with active urine sediment/proteinuria -> intrinsic renal disease becomes more likely than isolated prerenal change.

Check Next: eGFR -> urinalysis/ACR -> electrolytes -> repeat trend -> medication/volume review -> ultrasound if obstruction suspected.

Important Note: A stable creatinine can still represent important kidney disease in low-muscle-mass patients.

eGFR

About the Test: Estimated GFR translates creatinine (and sometimes cystatin C) into an estimate of kidney filtration. It is used for CKD staging but is less reliable during rapidly changing kidney function.

Reference Range: ≥ 90 mL/min/1.73 m² is generally considered normal/high when no other kidney-damage marker is present; CKD categories: G1 ≥ 90 , G2 60-89, G3a 45-59, G3b 30-44, G4 15-29, G5 < 15 .

↑ Increased	↓ Decreased
Term: Reduced eGFR Acute or chronic reduction in kidney filtration.	High eGFR is usually not a disease; hyperfiltration may occur in selected states.

Pattern Helps

- eGFR < 60 persisting ≥ 3 months $\rightarrow$ CKD criterion.
- eGFR reduced + urine ACR elevated $\rightarrow$ higher kidney and cardiovascular risk.

Clinical Correlation

eGFR persistently < 60 mL/min/1.73m² for ≥ 3 months $\rightarrow$ chronic kidney disease is present when criteria are met.

Low eGFR + albuminuria $\rightarrow$ higher CKD risk and stronger evidence of chronic kidney damage.

Acute creatinine change makes eGFR less reliable for staging; assess for AKI using serial creatinine and urine output.

Check Next: Urine ACR $\rightarrow$ urinalysis $\rightarrow$ repeat kidney function $\rightarrow$ BP/diabetes assessment; cystatin C when confirmation is appropriate.

Important Note: Do not stage CKD from a single low eGFR. eGFR is not reliable in rapidly changing creatinine.

Uric Acid

About the Test: Uric acid is the end product of purine metabolism. Levels reflect production, diet, cell turnover and renal excretion. Hyperuricemia increases risk of gout but does not diagnose gout by itself.

Reference Range: Men about 3.4-7.0 mg/dL; women about 2.4-6.0 mg/dL, method dependent.

↑ Increased	↓ Decreased
Term: Hyperuricemia Gout risk; reduced renal excretion/CKD; high cell turnover; tumor lysis; diuretics; alcohol/purine intake.	Term: Hypouricemia Uricosuric drugs; SIADH in some cases; renal tubular disorders; low production.

Pattern Helps

- Uric acid high + acute monoarthritis -> gout possible; crystal identification is more definitive.
- Uric acid high + K/phosphate/LDH high after chemotherapy -> tumor lysis syndrome pattern.

Clinical Correlation

Hyperuricemia + acute monoarthritis + urate crystals -> gout is strongly supported.

Hyperuricemia + tumor lysis context + high potassium/phosphate + low calcium -> tumor lysis syndrome should be considered urgently.

Low uric acid + hyponatremia in euvolemia may be seen in SIADH, but is not diagnostic alone.

Check Next: Creatinine/eGFR -> joint fluid crystal analysis if acute arthritis -> medication/diet review.

Important Note: Serum uric acid may be normal during an acute gout attack.

Sodium

About the Test: Sodium is the major extracellular cation. Serum sodium primarily reflects water balance rather than total-body sodium. Interpretation requires volume status, serum osmolality and sometimes urine osmolality/sodium.

Reference Range: About 135-145 mmol/L.

↑ Increased	↓ Decreased
Term: Hypernatremia	Term: Hyponatremia

Water loss; diabetes insipidus; inadequate water intake; osmotic losses; hypertonic sodium administration.

Excess water relative to sodium: SIADH; diuretics; heart/liver failure; adrenal insufficiency; renal disease; GI losses with replacement; hyperglycemia-related translocation.

Pattern Helps

- Na low + serum osmolality low + urine osmolality high -> impaired water excretion pattern, including SIADH depending on volume/endocrine context.
- Na low + glucose very high -> consider translocational hyponatremia; correct sodium for hyperglycemia.
- Na high + dilute urine -> consider diabetes insipidus if clinically appropriate.

Clinical Correlation

Sodium low + low serum osmolality + concentrated urine + urine sodium not suppressed in a euvoletic patient -> SIADH becomes likely after excluding adrenal/thyroid causes.

Sodium low + very high glucose -> translocational hyponatremia should be considered and corrected sodium assessed.

Sodium high + high serum osmolality + dilute urine -> diabetes insipidus or impaired concentrating ability should be considered.

Check Next: Serum osmolality -> glucose -> urine osmolality/sodium -> volume assessment -> cortisol/TSH when indicated.

Important Note: Rapid correction of chronic sodium disorders can be dangerous; severe symptomatic abnormalities require urgent clinical management.

Potassium

About the Test: Potassium is the major intracellular cation and is critical for cardiac and neuromuscular function. Abnormal values may be life-threatening, and specimen hemolysis can cause pseudohyperkalemia.

Reference Range: About 3.5-5.0 mmol/L.

↑ Increased	↓ Decreased
Term: Hyperkalemia Kidney failure; hypoaldosteronism; acidosis; tissue breakdown; medications; pseudohyperkalemia from hemolysis/thrombocytosis.	Term: Hypokalemia GI loss; diuretics; hyperaldosteronism; insulin/beta-agonist shift; renal tubular losses; magnesium deficiency.

Pattern Helps

- K high + creatinine high + bicarbonate low -> renal failure/acidosis pattern.
- K low + Mg low -> potassium may be refractory until magnesium is corrected.
- Unexpected K high + hemolysis index high -> pseudohyperkalemia possible.

Clinical Correlation

Potassium high + creatinine high + bicarbonate low -> renal failure/acid-base disturbance is a likely contributor.

Potassium low + metabolic alkalosis -> GI loss, diuretics or mineralocorticoid excess should be considered.

Unexpected high potassium + hemolyzed sample/normal ECG -> pseudohyperkalemia must be excluded before interpretation.

Check Next: Repeat non-hemolyzed sample if unexpected -> ECG for significant abnormalities -> creatinine/eGFR -> bicarbonate -> Mg -> medication review.

Important Note: Critical potassium values require prompt verification/communication according to laboratory policy.

Chloride

About the Test: Chloride is the major extracellular anion and helps maintain electroneutrality. It is interpreted mainly with sodium, bicarbonate and acid-base status.

Reference Range: About 98-107 mmol/L.

↑ Increased	↓ Decreased
Term: Hyperchloremia	Term: Hypochloremia

Normal-anion-gap metabolic acidosis;
saline load; dehydration; renal tubular
acidosis.

Vomiting/gastric loss; metabolic
alkalosis; diuretics; dilutional states.

Pattern Helps

- Chloride high + bicarbonate low + normal anion gap -> hyperchloremic metabolic acidosis.
- Chloride low + bicarbonate high -> metabolic alkalosis pattern, often from vomiting/diuretics.

Clinical Correlation

Chloride high + bicarbonate low + normal anion gap -> hyperchloremic metabolic acidosis is suggested (e.g., diarrhea or renal tubular acidosis).

Chloride low + bicarbonate high -> vomiting/NG loss or chloride-depletion metabolic alkalosis is likely.

Interpret chloride with sodium, bicarbonate and anion gap rather than in isolation.

Check Next: Bicarbonate -> anion gap -> sodium/potassium -> blood gas if significant.

Important Note: Chloride is usually most informative as part of an electrolyte and acid-base pattern.

Bicarbonate / Total CO₂

About the Test: Serum total CO₂ largely reflects bicarbonate and is a screening marker of metabolic acid-base status. Blood gas analysis is more definitive when significant acid-base disturbance is suspected.

Reference Range: About 22-29 mmol/L.

↑ Increased	↓ Decreased
Term: Elevated bicarbonate Metabolic alkalosis or compensation for chronic respiratory acidosis.	Term: Low bicarbonate Metabolic acidosis or compensation for respiratory alkalosis.

Pattern Helps

- Bicarbonate low + anion gap high -> high-anion-gap metabolic acidosis.
- Bicarbonate low + chloride high + normal anion gap -> hyperchloremic metabolic acidosis.

Clinical Correlation

Bicarbonate low + high anion gap -> high-anion-gap metabolic acidosis; check lactate, ketones, renal failure and toxins.

Bicarbonate low + normal anion gap + chloride high -> diarrhea or renal tubular acidosis should be considered.

Bicarbonate high + low chloride/potassium -> metabolic alkalosis from vomiting/diuretics is common.

Check Next: Calculate anion gap -> blood gas -> lactate/ketones/renal function/toxin evaluation according to pattern.

Important Note: Sample handling can affect total CO₂; correlate unexpected values with blood gas when necessary.

Calcium

About the Test: Calcium is essential for neuromuscular and bone function. Total calcium is influenced by albumin; ionized calcium is the biologically active fraction and is preferred when protein or acid-base abnormalities are significant.

Reference Range: Total calcium about 8.5-10.5 mg/dL; ionized calcium about 1.12-1.32 mmol/L, method dependent.

↑ Increased	↓ Decreased
Term: Hypercalcemia Primary hyperparathyroidism; malignancy; vitamin D excess/granulomatous disease; drugs; immobilization.	Term: Hypocalcemia Vitamin D deficiency; hypoparathyroidism; CKD; pancreatitis; hypomagnesemia; massive transfusion.

Pattern Helps

- Calcium high + PTH high/inappropriately normal -> PTH-mediated hypercalcemia, usually primary hyperparathyroidism.

- Calcium high + PTH suppressed -> malignancy/vitamin D-mediated and other non-PTH causes.
- Calcium low + phosphate high + CKD -> CKD-mineral bone disorder pattern.

Clinical Correlation

Calcium high + PTH high/inappropriately normal -> primary hyperparathyroidism is likely.

Calcium high + PTH suppressed -> malignancy, vitamin D excess/granulomatous disease or other non-PTH causes should be considered.

Calcium low + PTH high + vitamin D low -> vitamin D deficiency/secondary hyperparathyroidism is likely; use corrected or ionized calcium when albumin is abnormal.

Check Next: Albumin or ionized calcium -> PTH -> phosphate -> Mg -> 25-OH vitamin D -> creatinine/eGFR.

Important Note: Albumin-corrected calcium formulas have limitations; ionized calcium is preferred when accuracy matters.

Phosphate

About the Test: Phosphate is regulated by kidney function, PTH, FGF23 and vitamin D and is closely linked to bone and energy metabolism.

Reference Range: About 2.5-4.5 mg/dL in adults; higher in children.

↑ Increased	↓ Decreased
Term: Hyperphosphatemia CKD; hypoparathyroidism; cell lysis; phosphate load.	Term: Hypophosphatemia Refeeding; hyperparathyroidism; vitamin D deficiency; renal wasting; alcoholism; respiratory alkalosis.

Pattern Helps

- Phosphate high + calcium low + creatinine high -> advanced CKD pattern.
- Phosphate low + calcium high + PTH high -> primary hyperparathyroidism pattern.

Clinical Correlation

Phosphate high + creatinine/eGFR abnormal + calcium low/normal -> CKD-related mineral disorder is likely.

Phosphate low + PTH high -> renal phosphate wasting from hyperparathyroidism is possible.

Phosphate low + refeeding/alcoholism/insulin treatment -> intracellular shift should be considered.

Check Next: Calcium -> PTH -> vitamin D -> Mg -> kidney function; urine phosphate when renal wasting suspected.

Important Note: Severe hypophosphatemia can cause muscle weakness, respiratory failure and hemolysis.

Magnesium

About the Test: Magnesium is important for neuromuscular function and for potassium and calcium homeostasis. Deficiency often coexists with refractory hypokalemia or hypocalcemia.

Reference Range: About 1.7-2.4 mg/dL (0.70-1.00 mmol/L).

↑ Increased	↓ Decreased
Term: Hypermagnesemia Renal failure; excessive magnesium administration/antacids/laxatives.	Term: Hypomagnesemia GI loss; diuretics; alcoholism; PPIs; renal wasting; malnutrition.

Pattern Helps

- Mg low + K low -> potassium may remain low until Mg is corrected.
- Mg low + calcium low/PTH impairment -> hypocalcemia may be magnesium related.

Clinical Correlation

Magnesium low + refractory hypokalemia and/or hypocalcemia -> magnesium deficiency is likely contributing and should be corrected.

Magnesium high + renal failure -> impaired renal excretion is likely.

Magnesium low in chronic diarrhea, diuretic/PPI use or alcoholism is common and may cause arrhythmia/neuromuscular symptoms.

Check Next: Potassium -> calcium -> creatinine/eGFR -> medication/GI loss review.

Important Note: Serum magnesium may not perfectly reflect total-body stores, but marked abnormalities are clinically important.

5. Glucose & Metabolic

Quick-reference interpretation. Use each result as a starting point for correlation, not as a stand-alone diagnosis.

Fasting / Random Plasma Glucose

About the Test: Plasma glucose reflects current glycemia. Interpretation depends on fasting status, symptoms and whether testing is for screening, diagnosis or acute management.

Reference Range: Fasting plasma glucose commonly ~70-99 mg/dL. Diabetes diagnostic threshold: fasting ≥ 126 mg/dL or random ≥ 200 mg/dL with classic symptoms; diagnosis usually requires confirmation unless unequivocal.

↑ Increased	↓ Decreased
Term: Hyperglycemia Diabetes mellitus; stress illness; corticosteroids; endocrine disorders; pancreatitis/pancreatic disease.	Term: Hypoglycemia Insulin/sulfonylurea effect; fasting; severe liver disease; sepsis; adrenal insufficiency; insulinoma and other less common causes.

Pattern Helps

- Glucose high + HbA1c high -> chronic hyperglycemia/diabetes pattern.
- Glucose high + ketones high + bicarbonate low + anion gap high -> diabetic ketoacidosis pattern.
- Glucose low + inappropriately high insulin/C-peptide -> endogenous hyperinsulinism or secretagogue exposure depending on drug screen.

Clinical Correlation

Fasting glucose ≥ 126 mg/dL or random glucose ≥ 200 mg/dL with classic symptoms meets diabetes diagnostic criteria when guideline conditions are satisfied.

Glucose high + ketones high + bicarbonate low/high anion gap -> diabetic ketoacidosis should be considered urgently.

Low glucose + symptoms resolving after glucose correction -> hypoglycemia is clinically significant; investigate insulin/C-peptide/medication causes when unexplained.

Check Next: HbA1c -> repeat diagnostic testing as needed -> ketones/electrolytes if acute -> insulin/C-peptide only in selected hypoglycemia work-up.

Important Note: Use plasma-based diagnostic criteria; point-of-care meters are not generally used to establish diagnosis.

HbA1c

About the Test: HbA1c reflects average glycemic exposure over approximately 2-3 months and is used for diabetes diagnosis and monitoring. It is affected by red-cell lifespan, transfusion, hemoglobin variants and some anemias.

Reference Range: Normal commonly <5.7%; prediabetes 5.7-6.4%; diabetes diagnostic threshold >=6.5% using standardized laboratory methods.

↑ Increased	↓ Decreased
Term: Elevated HbA1c Chronic hyperglycemia/diabetes; values may be falsely high in some low-turnover RBC states.	Term: Low HbA1c Lower average glucose or falsely low result with shortened RBC survival, hemolysis, recent blood loss/transfusion.

Pattern Helps

- Glucose high + HbA1c high -> sustained hyperglycemia.
- Glucose high + HbA1c unexpectedly normal/low -> consider recent-onset hyperglycemia or HbA1c interference/altered RBC survival.

Clinical Correlation

HbA1c >=6.5% supports diabetes when diagnostic criteria and confirmation requirements are met.

High HbA1c + normal current glucose may reflect prior chronic hyperglycemia; discordance should prompt review of RBC lifespan/assay interference.

Unexpectedly low HbA1c + anemia/hemolysis/transfusion -> shortened RBC survival may falsely lower HbA1c.

Check Next: Fasting/random glucose -> repeat HbA1c or alternative glycemic marker if discordant; CBC/hemolysis review when interference suspected.

Important Note: HbA1c should not be relied on when RBC turnover or hemoglobin variants make it unreliable.

Serum beta-Hydroxybutyrate / Ketones

About the Test: Ketones are produced during increased fat metabolism. Blood beta-hydroxybutyrate is preferred for assessing diabetic ketoacidosis because urine nitroprusside tests do not directly measure beta-hydroxybutyrate.

Reference Range: Blood beta-hydroxybutyrate is usually <0.6 mmol/L; method/clinical thresholds vary.

↑ Increased	↓ Decreased
Term: Ketonemia / ketonuria Diabetic ketoacidosis; starvation/fasting; alcoholic ketoacidosis; low-carbohydrate states; vomiting.	Absent/low ketones are expected normally.

Pattern Helps

- Glucose high + ketones high + bicarbonate low + anion gap high -> DKA strongly suspected.
- Ketones high + glucose normal/low -> starvation, alcoholic or euglycemic ketoacidosis should be considered.

Clinical Correlation

Ketones high + hyperglycemia + high-anion-gap metabolic acidosis -> DKA is strongly suspected.

Ketones high + normal/low glucose -> starvation, alcohol-related ketosis or euglycemic DKA should be considered in the right context.

Beta-hydroxybutyrate is preferred over urine ketones for assessing active ketoacidosis.

Check Next: Glucose -> electrolytes/bicarbonate -> anion gap -> venous/arterial blood gas -> lactate as indicated.

Important Note: Urine ketones can lag behind clinical change and may paradoxically rise during early DKA treatment.

Lactate

About the Test: Lactate rises when production exceeds clearance. Elevation can reflect tissue hypoperfusion, adrenergic stress, seizures, drugs/toxins or impaired hepatic clearance.

Reference Range: About 0.5-2.2 mmol/L in venous plasma, method and sampling dependent.

↑ Increased	↓ Decreased
Term: Hyperlactatemia / lactic acidosis when associated with acidemia Shock/sepsis; hypoxia/hypoperfusion; seizures; severe asthma/beta-agonists; liver dysfunction; metformin-associated states; toxins; mitochondrial disorders.	Low lactate is not clinically significant.

Pattern Helps

- Lactate high + low pH + high anion gap -> lactic acidosis pattern.
- Serial falling lactate in shock can support improving perfusion, interpreted with the whole clinical picture.

Clinical Correlation

Lactate high + hypotension/organ dysfunction -> tissue hypoperfusion/sepsis or shock should be considered urgently.

Lactate high without hypotension -> seizures, beta-agonists, severe hepatic dysfunction, drugs/toxins or impaired clearance may contribute.

Persistently rising lactate despite treatment is concerning and requires reassessment of perfusion/source.

Check Next: Blood gas -> anion gap -> renal/liver function -> sepsis/shock evaluation -> medication/toxin review.

Important Note: Tourniquet time, fist clenching and delayed processing can affect lactate. Clinical urgency depends on context and degree of elevation.

6. Lipid Profile

Quick-reference interpretation. Use each result as a starting point for correlation, not as a stand-alone diagnosis.

Total Cholesterol

About the Test: Total cholesterol measures cholesterol carried by all major lipoprotein particles. Cardiovascular risk assessment relies more heavily on LDL-C, non-HDL-C and overall clinical risk than on total cholesterol alone.

Reference Range: Desirable generally <200 mg/dL.

↑ Increased	↓ Decreased
Term: Hypercholesterolemia Primary dyslipidemia; hypothyroidism; nephrotic syndrome; cholestasis; diabetes/obesity; medications.	Term: Low cholesterol Malnutrition; hyperthyroidism; chronic inflammatory illness; severe liver disease.

Pattern Helps

- Total cholesterol high + LDL-C high -> atherogenic hypercholesterolemia.

Clinical Correlation

High total cholesterol + high LDL-C -> atherogenic hypercholesterolemia is likely and cardiovascular risk assessment is needed.

High total cholesterol with very high HDL-C may not reflect the same risk as LDL-driven elevation; interpret components, not total alone.

Low cholesterol can occur in severe illness, malnutrition, hyperthyroidism or advanced liver disease.

Check Next: LDL-C -> HDL-C -> triglycerides -> non-HDL-C -> secondary-cause evaluation (TSH, glucose, renal/liver tests) when appropriate.

Important Note: Treatment decisions should be based on cardiovascular risk and guideline thresholds, not total cholesterol alone.

LDL Cholesterol

About the Test: LDL-C estimates cholesterol within low-density lipoproteins and is a major atherogenic marker used in cardiovascular prevention.

Reference Range: Optimal commonly <100 mg/dL; treatment targets/thresholds depend on cardiovascular risk.

↑ Increased	↓ Decreased
Term: Elevated LDL-C Primary/familial hypercholesterolemia; hypothyroidism; nephrotic syndrome; cholestasis; diet/metabolic factors.	Term: Low LDL-C May reflect treatment, hyperthyroidism, malnutrition, chronic illness or rare genetic states.

Pattern Helps

- LDL-C very high (e.g., ≥190 mg/dL) -> consider severe primary hypercholesterolemia/familial hypercholesterolemia after secondary causes are excluded.

Clinical Correlation

High LDL-C with family history/very high levels -> familial hypercholesterolemia should be considered.

High LDL-C + diabetes/CKD/ASCVD -> overall cardiovascular risk is high and treatment intensity depends on guideline context.

Calculated LDL-C may be unreliable with very high triglycerides; use an appropriate direct/alternative method when needed.

Check Next: Repeat fasting/nonfasting lipid panel as appropriate -> secondary cause testing -> family history; ApoB/Lp(a) in selected risk assessment.

Important Note: Calculated LDL-C becomes less reliable with marked hypertriglyceridemia; direct methods or alternative markers may be needed.

HDL Cholesterol

About the Test: HDL-C is inversely associated with cardiovascular risk at a population level, but very high HDL is not necessarily protective and HDL-C is not usually a direct treatment target.

Reference Range: Low HDL-C often <40 mg/dL in men and <50 mg/dL in women; ≥60 mg/dL has traditionally been considered high.

↑ Increased	↓ Decreased
High HDL-C is not a disease and should	Term: Low HDL-C

be interpreted with overall risk.

Insulin resistance/metabolic syndrome;
smoking; obesity; hypertriglyceridemia;
inactivity; genetic factors.

Pattern Helps

- Low HDL + high triglycerides + hyperglycemia/central obesity -> insulin resistance/metabolic syndrome pattern.

Clinical Correlation

Low HDL-C + high triglycerides + insulin resistance features -> atherogenic metabolic syndrome pattern is common.

HDL-C should not be interpreted as a stand-alone protective factor; LDL/non-HDL/ApoB and global risk remain central.

Extremely high HDL-C is not necessarily protective and may reflect genetic, alcohol or other factors.

Check Next: Full lipid panel -> glucose/HbA1c -> BP/clinical risk assessment.

Important Note: Do not interpret HDL-C in isolation as a measure of protection.

Triglycerides

About the Test: Triglycerides reflect triglyceride-rich lipoproteins and are strongly influenced by meals, alcohol, diabetes, obesity, drugs and genetic disorders. Severe elevation increases pancreatitis risk.

Reference Range: Normal fasting <150 mg/dL.

↑ Increased	↓ Decreased
Term: Hypertriglyceridemia Diabetes/insulin resistance; obesity; alcohol; hypothyroidism; CKD/nephrotic syndrome; drugs; familial disorders.	Term: Low triglycerides Usually not clinically significant; can occur with malnutrition or hyperthyroidism.

Pattern Helps

- Triglycerides ≥ 500 mg/dL -> pancreatitis risk rises; ≥ 1000 mg/dL is particularly concerning.

- High TG + low HDL + high glucose -> insulin resistance pattern.

Clinical Correlation

Triglycerides very high (especially ≥ 500 mg/dL, markedly so ≥ 1000) -> pancreatitis risk increases and secondary causes should be sought.

High triglycerides + low HDL + hyperglycemia/central obesity -> insulin resistance/metabolic syndrome is likely.

High triglycerides + abdominal pain + elevated lipase -> hypertriglyceridemic pancreatitis should be considered.

Check Next: Repeat fasting level if needed -> glucose/HbA1c -> TSH -> renal/liver tests -> medication/alcohol review.

Important Note: Marked lipemia can interfere with some laboratory assays.

7. Cardiac & Muscle Markers

Quick-reference interpretation. Use each result as a starting point for correlation, not as a stand-alone diagnosis.

High-Sensitivity Cardiac Troponin

About the Test: Cardiac troponin I or T is the preferred biomarker of myocardial injury. High-sensitivity assays detect small changes; myocardial infarction requires a rise/fall pattern plus clinical evidence of ischemia, not merely an elevated value.

Reference Range: Assay-specific 99th percentile upper reference limit; use the manufacturer/laboratory-specific cutoff, often sex-specific.

↑ Increased	↓ Decreased
Term: Myocardial injury Acute coronary syndrome/MI; myocarditis; heart failure; tachyarrhythmia; pulmonary embolism; sepsis; renal disease; strenuous exercise and other cardiac stress.	Low/undetectable troponin is expected in healthy individuals but interpretation depends on timing and assay.

Pattern Helps

- Troponin above 99th percentile + significant rise/fall -> acute myocardial injury.
- Rise/fall + ischemic symptoms/ECG/imaging evidence -> acute MI should be considered.
- Persistently elevated stable troponin -> chronic myocardial injury possible.

Clinical Correlation

Rise/fall in troponin above the assay 99th percentile + ischemic symptoms/ECG changes -> acute myocardial infarction should be considered.

Persistently elevated troponin without dynamic change -> chronic myocardial injury (e.g., CKD/structural heart disease) may be present.

Troponin elevation alone indicates myocardial injury, not automatically myocardial infarction.

Check Next: Serial troponin per validated pathway -> ECG -> clinical assessment -> renal function and other investigations according to context.

Important Note: Never diagnose MI from a single elevated troponin without the required clinical evidence.

CK / CK-MB

About the Test: Creatine kinase reflects skeletal and cardiac muscle injury. CK-MB has largely been replaced by troponin for myocardial infarction diagnosis but can still be useful in selected contexts.

Reference Range: Total CK often ~30-200 U/L, strongly sex/muscle/method dependent; CK-MB assay-specific.

↑ Increased	↓ Decreased
Term: Elevated CK Rhabdomyolysis; muscle trauma; strenuous exercise; myositis; seizures; intramuscular injections; drugs such as statins.	Low CK is usually not clinically important.

Pattern Helps

- CK markedly high + AST high -> skeletal muscle injury/rhabdomyolysis likely.
- CK high + creatinine rising + K/phosphate high -> rhabdomyolysis with kidney injury pattern.

Clinical Correlation

CK markedly high + muscle pain/weakness + urine heme with few RBCs -> rhabdomyolysis is strongly suspected.

CK-MB rise with troponin rise may support myocardial injury, but troponin is the preferred cardiac biomarker.

CK elevation after exercise, trauma, seizures or injections may be non-cardiac.

Check Next: Creatinine/eGFR -> potassium/phosphate/calcium -> urinalysis (heme positive with few RBCs may indicate myoglobin) -> medication/trauma review.

Important Note: CK-MB can rise from skeletal muscle and is less specific than troponin.

BNP / NT-proBNP

About the Test: BNP and NT-proBNP are released in response to myocardial wall stress and support diagnosis/risk assessment in heart failure. Concentrations are influenced by age, kidney function, atrial fibrillation and obesity.

Reference Range: Decision thresholds vary by assay, age and acute vs non-acute setting; use validated laboratory/clinical cutoffs.

↑ Increased	↓ Decreased
Term: Elevated natriuretic peptide Heart failure; renal dysfunction; atrial fibrillation; pulmonary hypertension/PE; advanced age; critical illness.	Low values make significant heart failure less likely in appropriate settings.

Pattern Helps

- Dyspnea + high BNP/NT-proBNP + congestion on imaging/exam -> heart failure more likely.

Clinical Correlation

BNP/NT-proBNP high + dyspnea + imaging/clinical congestion -> heart failure is strongly supported.

Low natriuretic peptide in an appropriate acute dyspnea pathway makes heart failure less likely.

Levels can be elevated in renal failure, older age and atrial fibrillation and lower in obesity; use assay/context-specific thresholds.

Check Next: Clinical assessment -> ECG -> echocardiography -> renal function -> troponin as indicated.

Important Note: Obesity can lower natriuretic peptide concentrations; renal dysfunction can raise them.